

Shourya Trikha

Junior Undergraduate
Department of Computer Science and Engineering

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ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	Bachelor of Technology	Indian Institute of Technology Kanpur	8.61 /10.0
2021	CBSE (XII)	Disha Delphi Public School, Kota	96.6%
2019	CBSE (X)	Maa Bharti Sr. Secondary School, Kota	95.2%

SCHOLASTIC ACHIEVEMENTS

- Recipient of the **Academic Excellence Award** for exceptional performance in the first year at IIT Kanpur
- Secured **All India Rank 206** in **Joint Entrance Examination - Advanced 2021** among **1.50 Lakh** candidates
- Secured **All India Rank 43** in **Joint Entrance Examination - Main 2021** among **10.5 Lakh** candidates
- Secured **100 percentile** in **Maths** and **Physics** with **100/100** score in **Joint Entrance Examination - Main 2021**
- Awarded with prestigious **KVPY-SA, 2019** fellowship, with **All India Rank 55** conducted by **IISc Bangalore**
- Attended the **Junior Science Orientation cum Selection Camp**, the final stage in India leading to **IJSO**
- Among the national top **1% in IOQP**, and qualified for **Indian National Physics Olympiad, 2021**
- Among the national top **1% in IOQC**, and qualified for **Indian National Chemistry Olympiad, 2021**

INTERNSHIP EXPERIENCE

SURGE IITK 📍 (Research Internship) (May '23 - July '23)

- Mentor:** Professor Ashutosh Modi (CSE, Indian Institute of Technology Kanpur)
- Implemented models such as CNNs, LSTMs and BiLSTMs to perform **Sentence Boundary detection** on Legal Texts
 - Used **SpaCy** and other deep learning models to perform SBD with **F1 Score 0.72 & 0.84** respectively (on US data)
 - Prepared an **annotated** dataset of **50+** Indian legal documents and used it for training and testing the SBD models

KEY PROJECTS

Even-Split 📍 (CS253 Course Project) (January '23 - April '23)

- Mentor:** Professor Indranil Saha (CSE, Indian Institute of Technology Kanpur)
- Developed a WebApp for **optimising expenses within a group** and a **unified calendar** for the whole campus
 - Designed the front-end with **Bootstrap** and **JavaScript** and back-end with **MongoDB Framework**
 - Implemented a sophisticated algorithm in **Javascript** to **minimise the number of transactions** within a group
 - Completed Requirement and Design document followed by implementation, **beta testing** and bug removal

Space Travel System 📍 (CS253 Course Project) (January '23 - April '23)

- Mentor:** Professor Indranil Saha (CSE, Indian Institute of Technology Kanpur)
- Designed an **OOPs** (in C++) based ticket booking system for Space travel through **Command Line Interface (CLI)**
 - Added functionality for booking, deleting, updating and viewing tickets, as well as add & remove Users and destinations
 - Incorporated **error and exception handling** to ensure robustness in handling unexpected inputs and user errors

CSE-BUBBLE 📍 (CS220 Course Project) (January '23 - April '23)

- Mentor:** Professor Urbi Chatterjee (CSE, Indian Institute of Technology Kanpur)
- Designed and implemented a single cycle processor with ISA similar to **MIPS-32**
 - Built the **Arithmetic Logic Unit (ALU)** and **Control Unit (CU)** in **top-down** approach to develop different **verilog** modules for different types of instructions and control the execution respectively
 - Developed the **MIPS code** for **Bubble Sort** and tested the processor by converting it into **machine code**

Cervical Cancer Screening 📍 (CS253 Course Project) (March '23 - March '23)

- Mentor:** Professor Indranil Saha (CSE, Indian Institute of Technology Kanpur)
- Built **SVM** and **KNN** based binary classifiers with accuracy **0.9 & 0.85** respectively, using Kaggle dataset
 - Utilized **scikit-learn**, **imblearn** and other libraries for data preprocessing, model training and result analysis
 - Employed **feature engineering** and fine tuned the hyperparameters for achieving higher accuracy

TECHNICAL SKILLS

- **Programming Skills:** C, C++, Python, JavaScript
- **Libraries:** Numpy, Pandas, Matplotlib, STL, Django
- **Utilities:** Git, Bash, Verilog HDL, MIPS, $\LaTeX$
- **Miscellaneous:** AutoCAD, Xilinx, Google Colab, Jupyter

RELEVANT COURSES

Software Development and Operations (CS253)	Probability and Statistics (CS203)	Discrete Mathematics (CS201)
Data Structures and Algorithms (ESO207)	Logic for Computer Science (CS202)	Computer Organisation (CS220)
Fundamentals of Computing (ESC101)	Linear Algebra and ODE (MTH102)	Real Analysis (MTH101)

POSITION OF RESPONSIBILITY

Organizer, Events and Competitions-Udghosh'23 (Annual Sports Festival, IIT Kanpur) (June '23 - Present)

- Leading a **4-tier** team of **80+** students responsible for organizing formal and informal events during the fest and pre-fest
- Organizing competitions like **UNOSQ**, **Chess**, **Esports** with **5000+** participation and **40,000+** eyeballs
- Launched **"Campus Ambassador Program"** in PAN India, receiving 5000+ participation and providing career internships

EXTRA-CURRICULAR

- Volunteered with Students' Placement Office, IIT Kanpur for smooth and timely conduction of Placement drive'22
- Initiated **Udaan** – a social initiative for specially-abled students with the help of the **CDAP** community, IIT Kanpur